ServiceNow Capstone Project

CAP-SN-001 | Evidence and Implementation Report

Candidate: Hari Kumar

Instance release: Brazil

Naming convention: CAP-GH

PART A – CUSTOMER AND CASE

A1 – Customer Setup

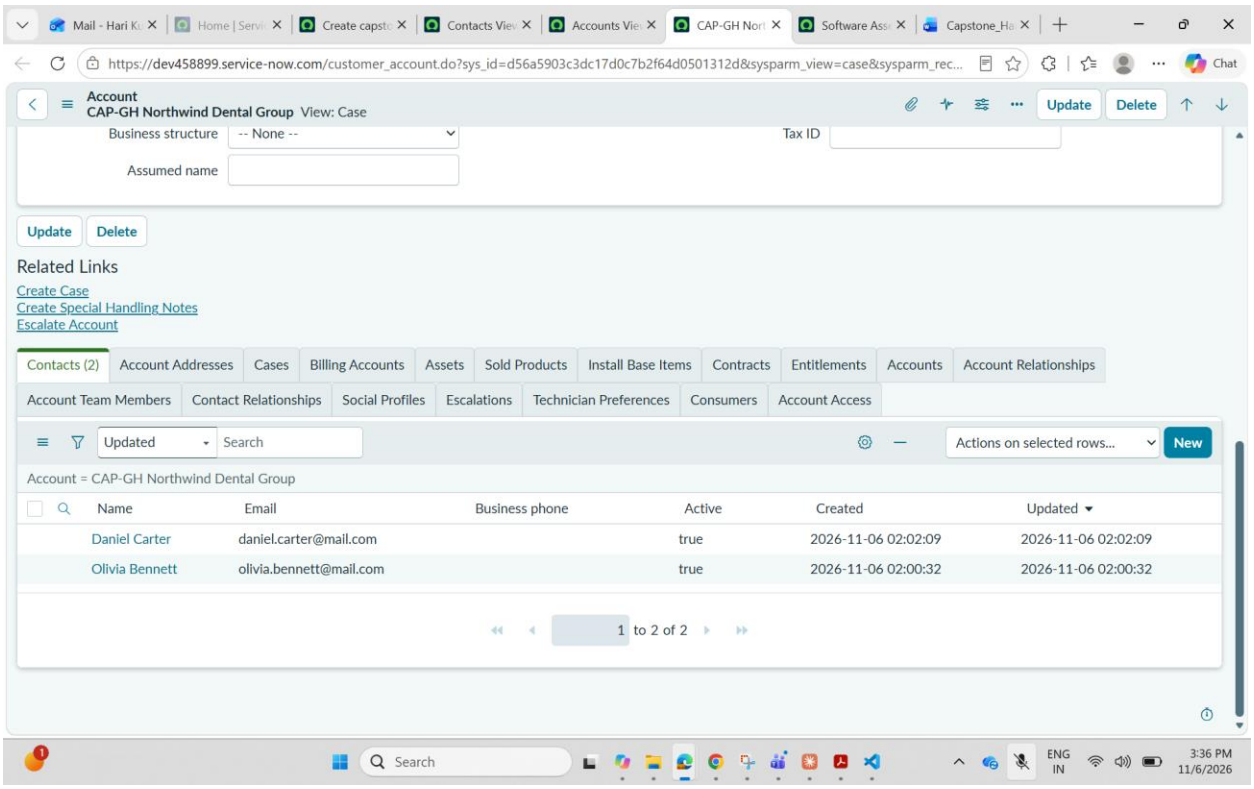
Objective

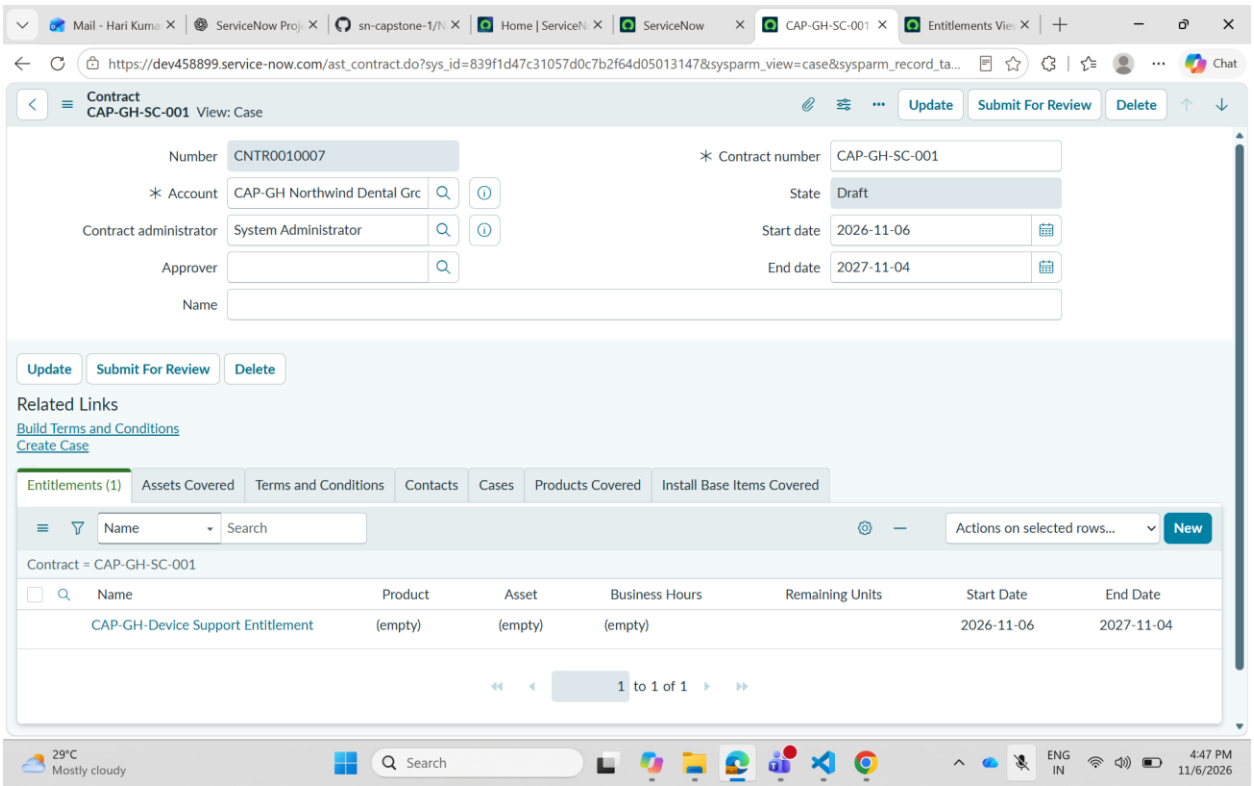
To configure the customer account, associate two contacts with different business roles, and establish an active service contract with a device support entitlement.

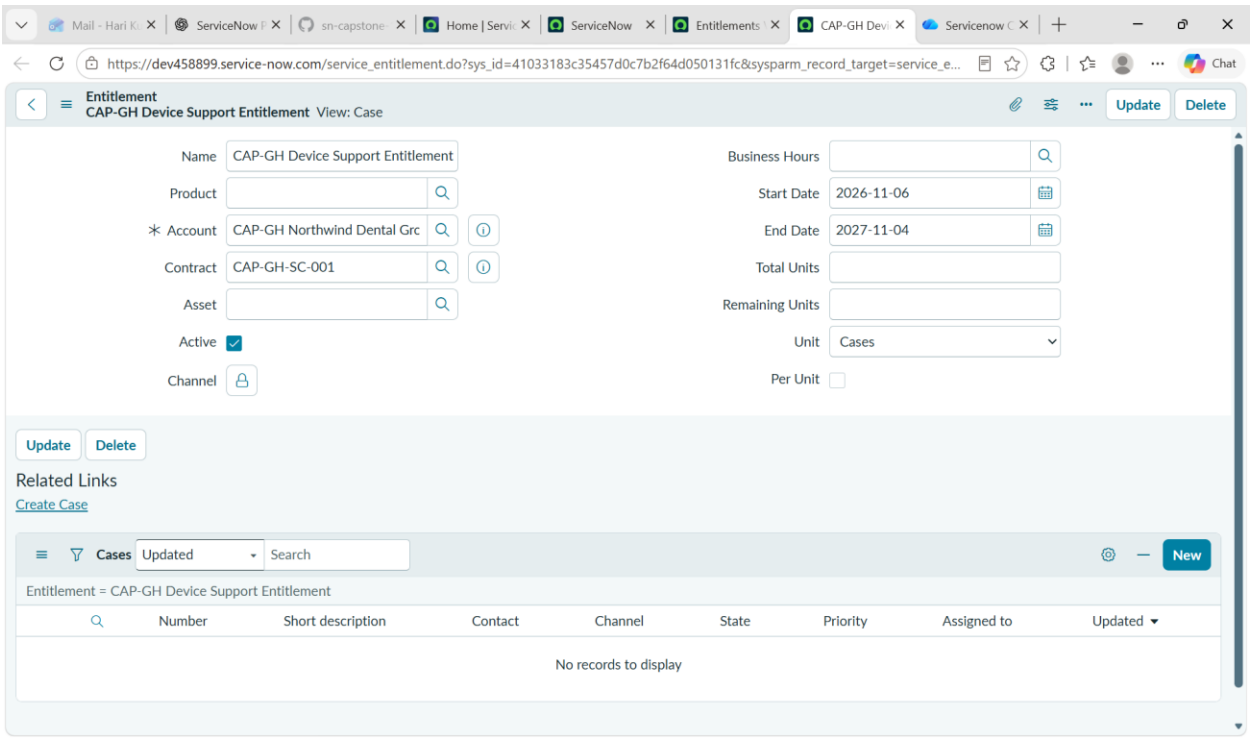
Implementation details

- Customer account: CAP-GH Northwind Dental Group
- Contact 1: Olivia Bennett – Practice Manager
- Contact 2: Daniel Carter – IT Lead
- Service contract: CAP-GH Northwind Dental Support Contract
- Entitlement: CAP-GH Device Support Entitlement

Evidence


https://dev458899.service-now.com/customer_account.do?sys_id=d56a5903c3dc17d0c7b2f64d0501312d&sysparm_view=case&sysparm_rec...
Account
CAP-GH Northwind Dental Group View: Case
Update
Delete
Business structure
-- None --
Tax ID
Assumed name
Update
Delete
Related Links
Create Case
Create Special Handling Notes
Escalate Account
Contacts (2)
Account Addresses
Cases
Billing Accounts
Assets
Sold Products
Install Base Items
Contracts
Entitlements
Accounts
Account Relationships
Account Team Members
Contact Relationships
Social Profiles
Escalations
Technician Preferences
Consumers
Account Access
Updated
Search
Actions on selected rows...
New
Account = CAP-GH Northwind Dental Group
Name
Email
Business phone
Active
Created
Updated
Daniel Carter
daniel.carter@mail.com
true
2026-11-06 02:02:09
2026-11-06 02:02:09
Olivia Bennett
olivia.bennett@mail.com
true
2026-11-06 02:00:32
2026-11-06 02:00:32
1 to 2 of 2
Search
ENG
IN
3:36 PM
11/6/2026


https://dev458899.service-now.com/ast_contract.do?sys_id=839f1d47c31057d0c7b2f64d05013147&sysparm_view=case&sysparm_record_ta...
Contract
CAP-GH-SC-001 View: Case
Update
Submit For Review
Delete
Number
CNTR0010007
Contract number
CAP-GH-SC-001
Account
CAP-GH Northwind Dental Grc
State
Draft
Contract administrator
System Administrator
Start date
2026-11-06
Approver
End date
2027-11-04
Name
Update
Submit For Review
Delete
Related Links
Build Terms and Conditions
Create Case
Entitlements (1)
Assets Covered
Terms and Conditions
Contacts
Cases
Products Covered
Install Base Items Covered
Name
Search
Actions on selected rows...
New
Contract = CAP-GH-SC-001
Name
Product
Asset
Business Hours
Remaining Units
Start Date
End Date
CAP-GH-Device Support Entitlement
(empty)
(empty)
(empty)
2026-11-06
2027-11-04
1 to 1 of 1
29°C
Mostly cloudy
Search
ENG
IN
4:47 PM
11/6/2026



A2 – Knowledge Article

Objective

To create and publish a knowledge article containing troubleshooting instructions for a laptop that fails to power on. The article will help customer service agents perform initial troubleshooting before escalating the issue.

Article Title

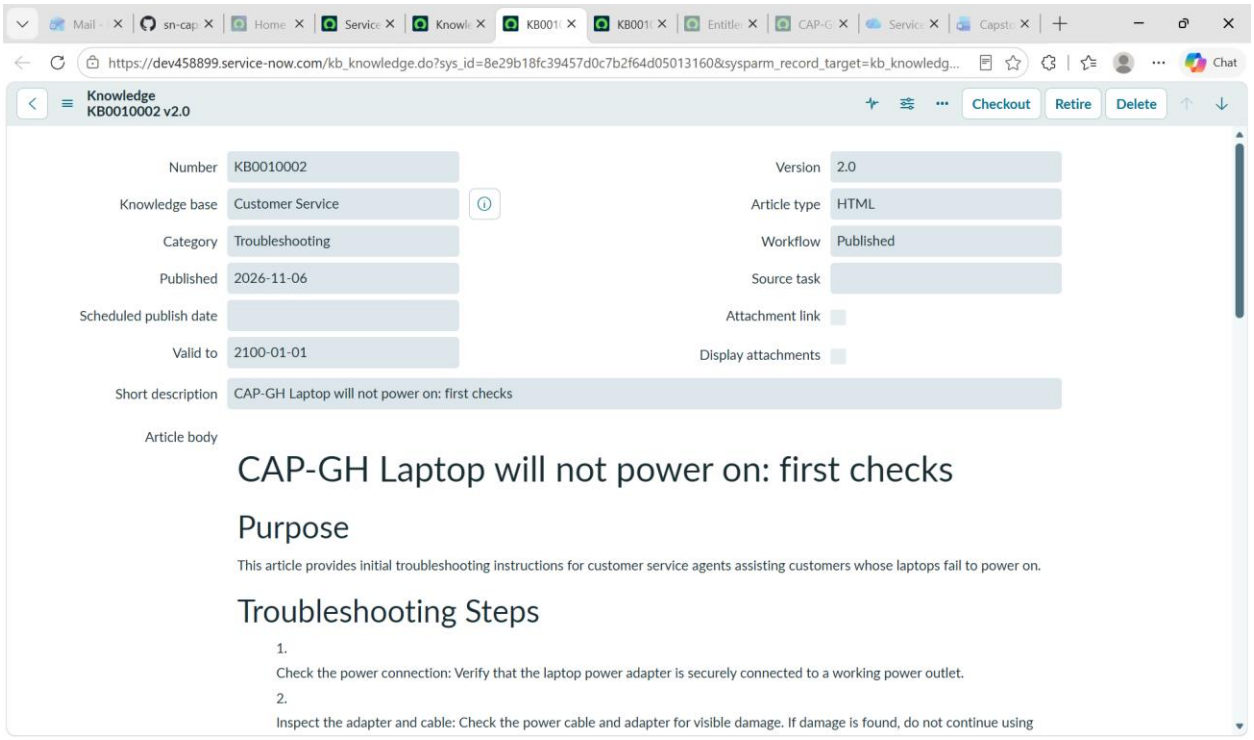
CAP-GH Laptop will not power on: first checks

Troubleshooting Content

1. Verify that the laptop power adapter is securely connected to a working power outlet.
2. Check the power cable and adapter for visible damage.
3. Disconnect external devices and hold the power button for approximately 15 seconds.
4. Reconnect the adapter and attempt to power on the laptop.

5. If the laptop still does not power on, arrange a technician inspection or replacement.

Evidence – A2-1



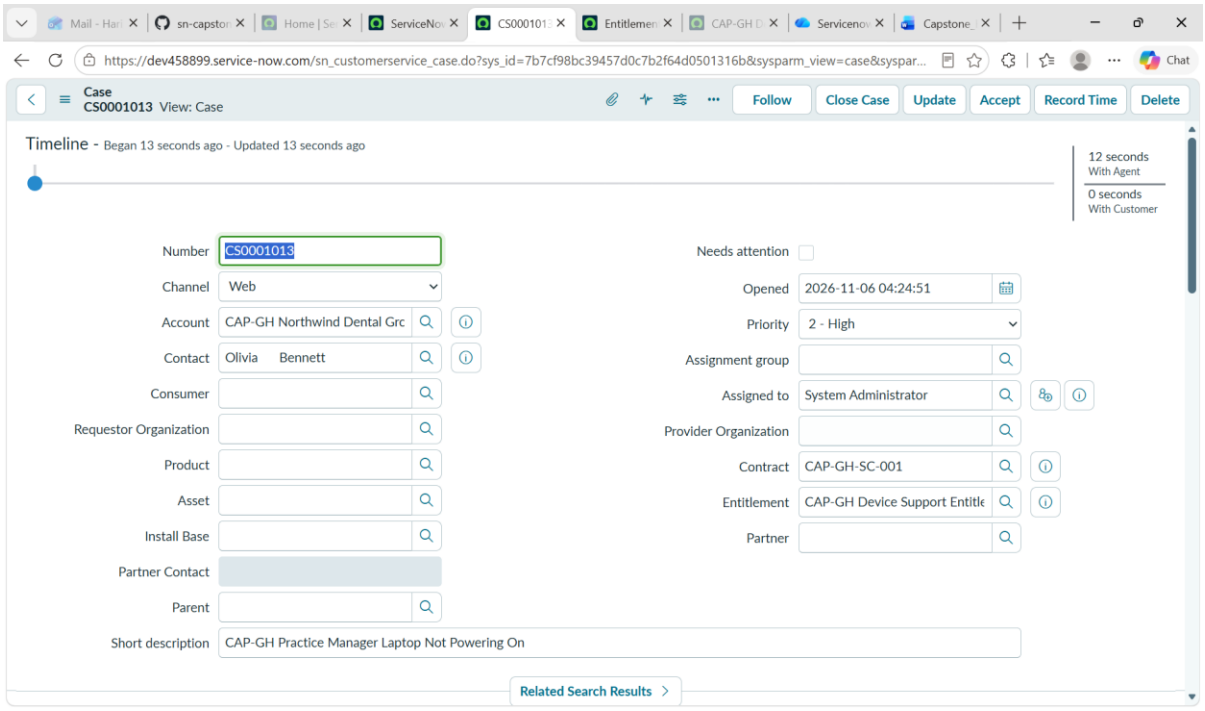
A3 – Case from First Contact to Resolution (10 Marks)

Objective:

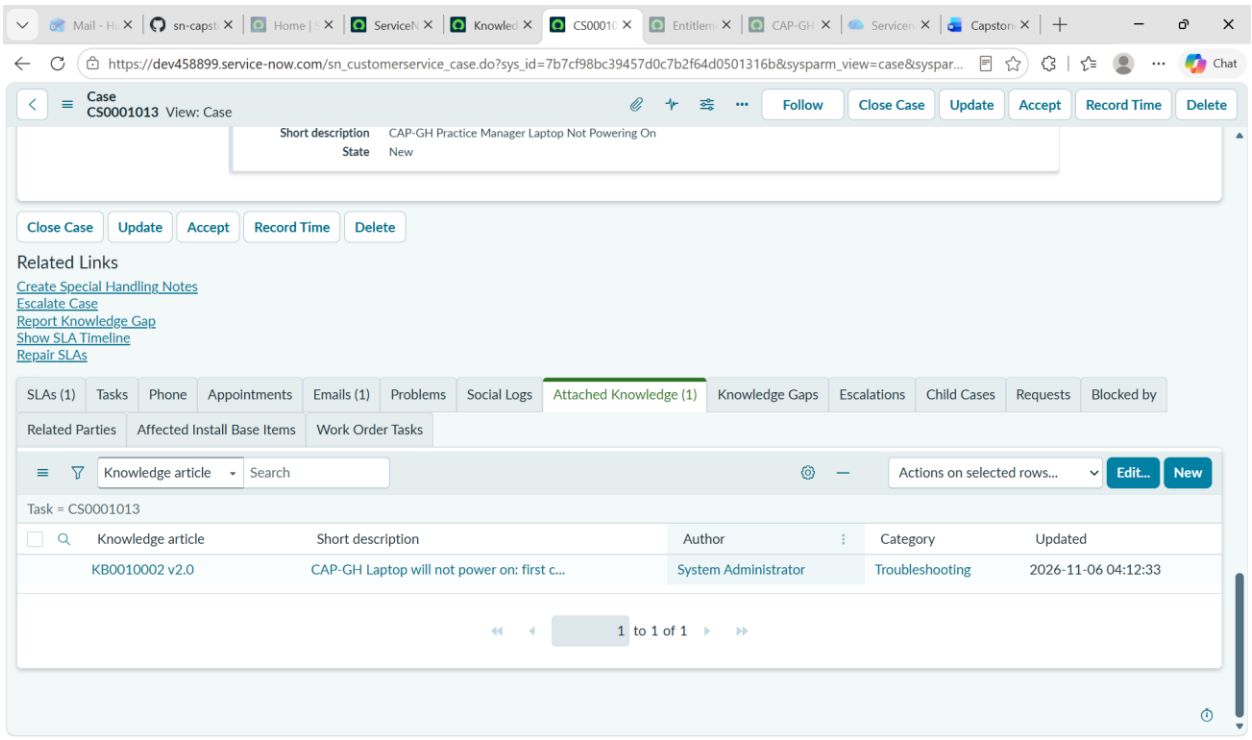
To create and manage a Customer Service Management (CSM) case for a laptop power failure reported by the Practice Manager of Northwind Dental Group. The case connects the customer account, contact, service contract and device support entitlement to ensure proper support tracking and resolution.

Evidence:

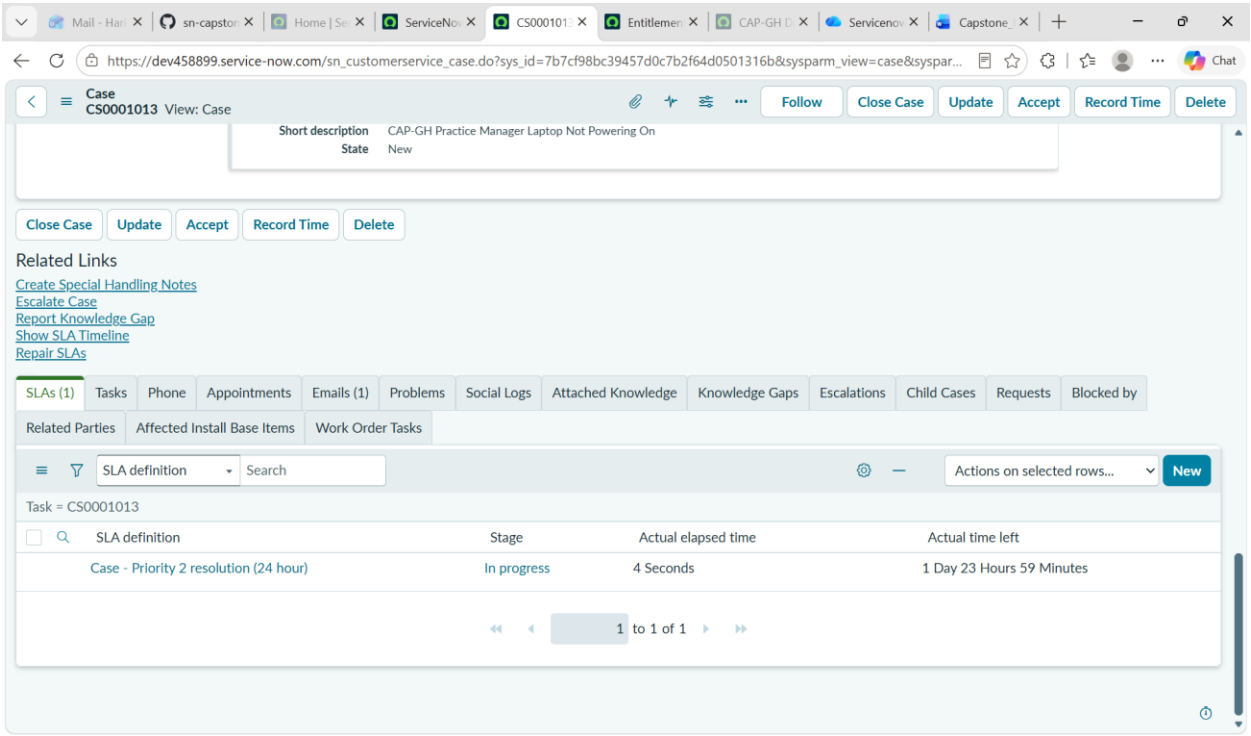
- **A3-1:** Customer service case showing the case number, account, contact and case details.



- **A3-2:** Case showing the associated troubleshooting knowledge article.



- **A3-3:** Case showing the attached Service Level Agreement (SLA).



- **A3-4:** Resolved case displaying the resolution notes after completion of field service activities.

A4 – Playbook

Objective:

To create a guided playbook that standardises the laptop replacement process and helps customer service agents coordinate customer communication and field technician activities.

Playbook Name:

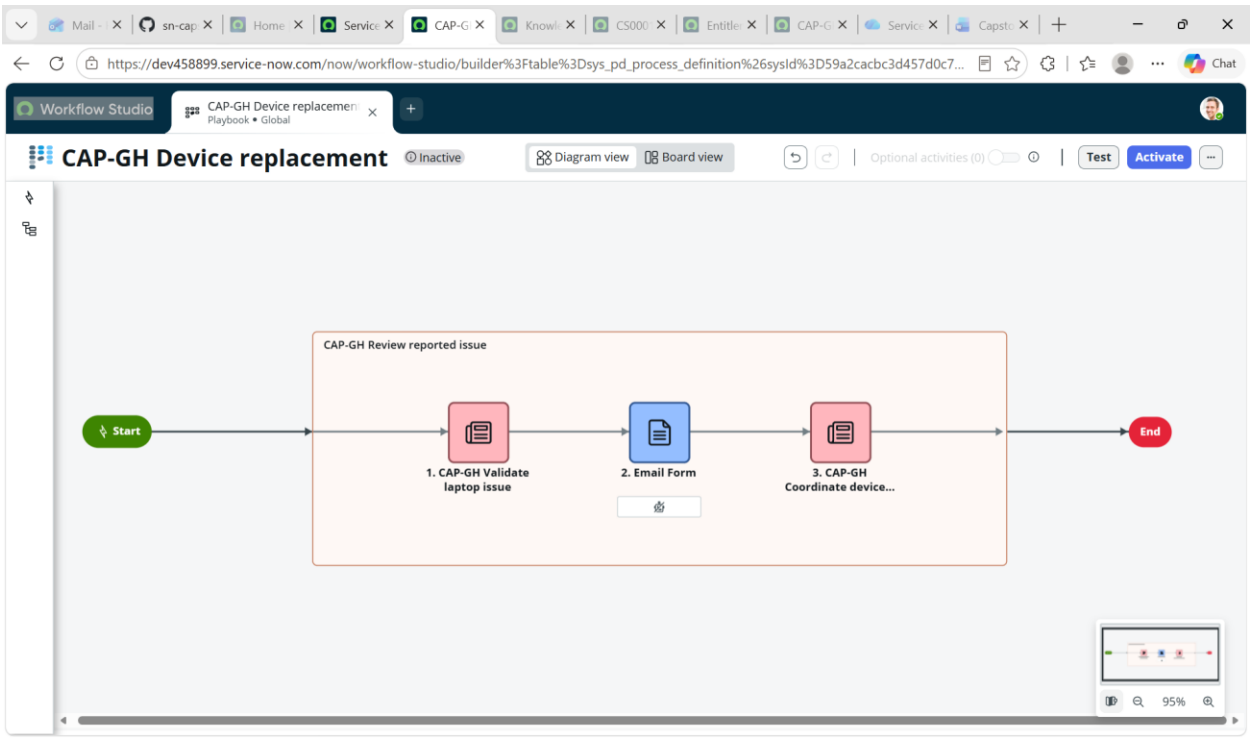
CAP-GH Device replacement

Planned Activities:

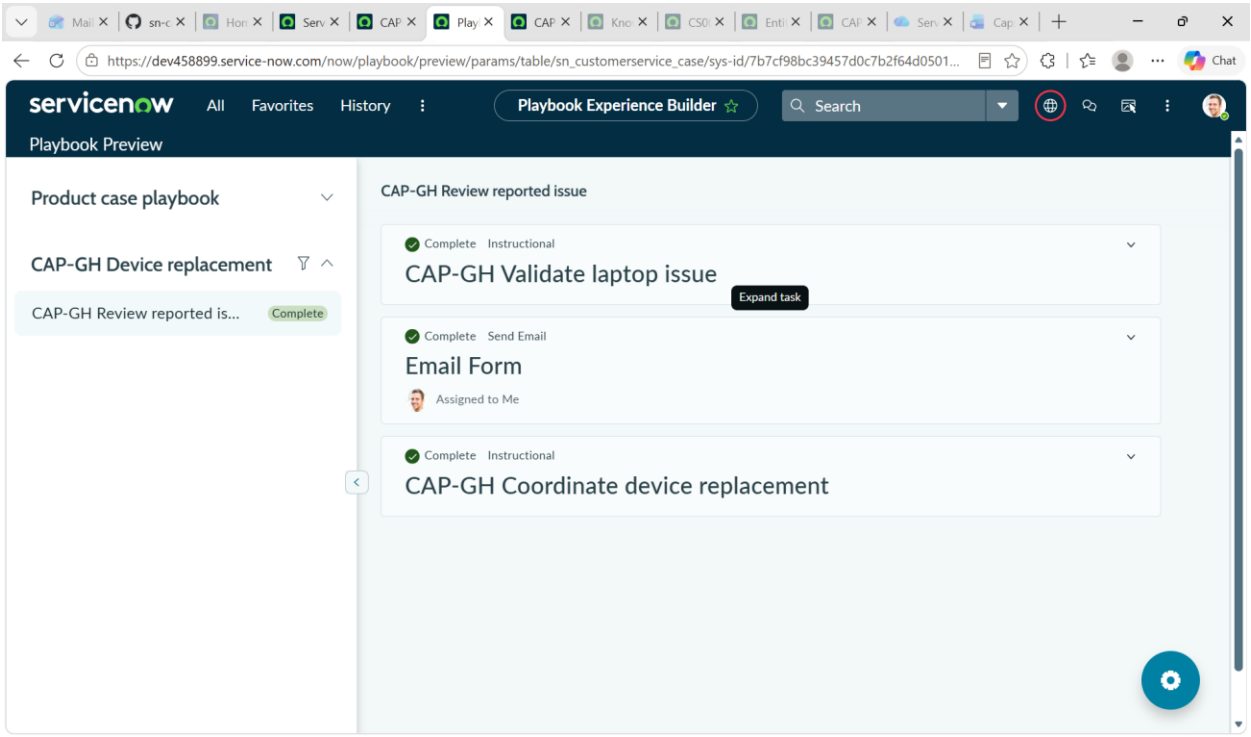
1. Review Laptop Failure – The agent reviews the customer case, verifies the initial troubleshooting information and confirms whether device replacement is necessary.
2. Customer Communication – The agent informs the customer about the replacement plan, explains the technician visit and confirms availability.
3. Coordinate Technician Visit – The agent coordinates the field technician visit, tracks the diagnosis and confirms completion of the replacement.

Evidence:

- A4-1: Playbook configuration showing its name and activities.



- A4-2: Published or activated playbook.



Outcome:

To be completed after implementation and verification.

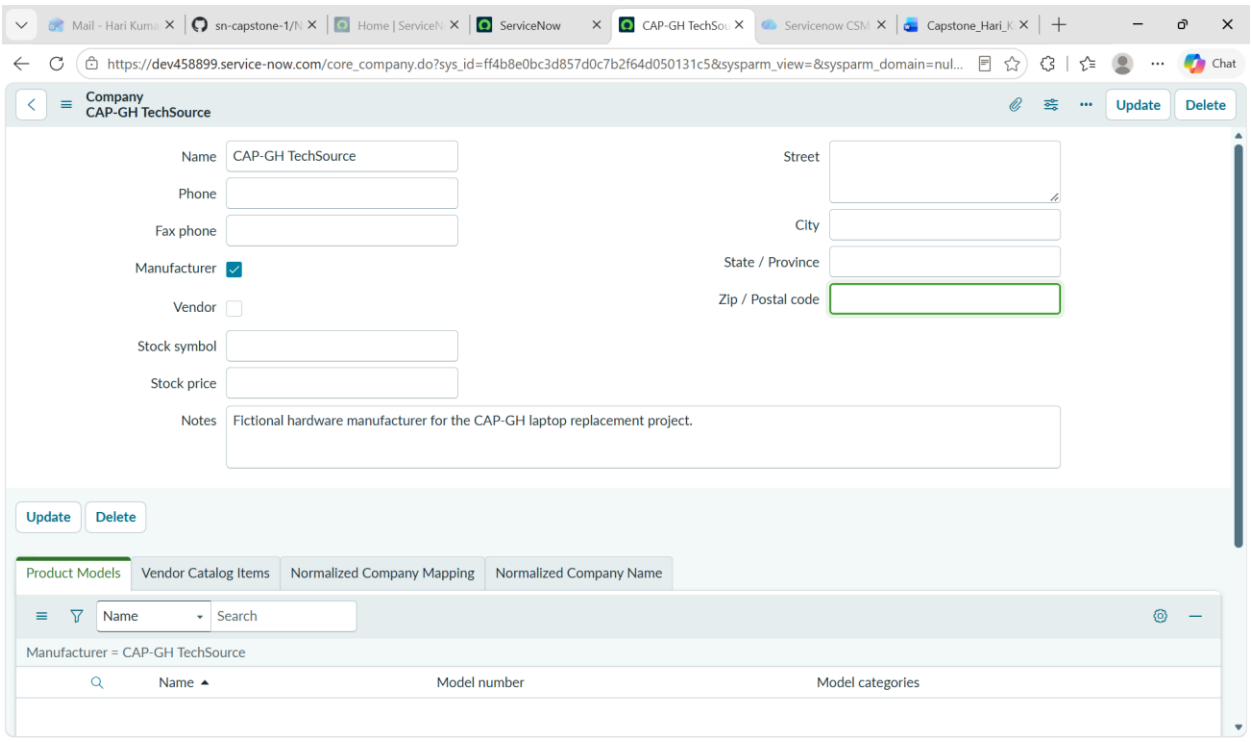
C1 – Master Data Configuration

Created the hardware manufacturer **CAP-GH TechSource** in ServiceNow. Configured the hardware model **CAP-GH Laptop 15**, associated it with the created manufacturer, and selected the **Computer** model category.

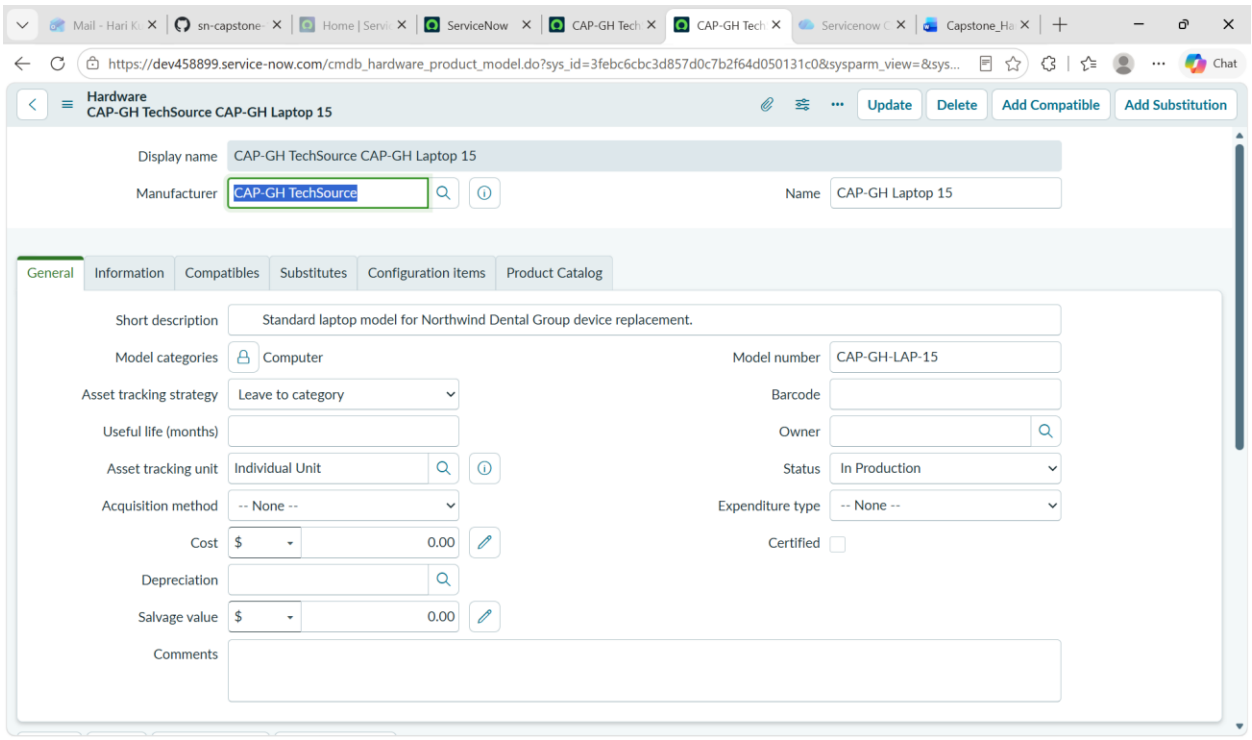
Also created the stockroom **CAP-GH Stockroom** and assigned an existing location and a stockroom manager. These master data records provide the foundation for managing the faulty and replacement laptops during the device replacement process.

Evidence:

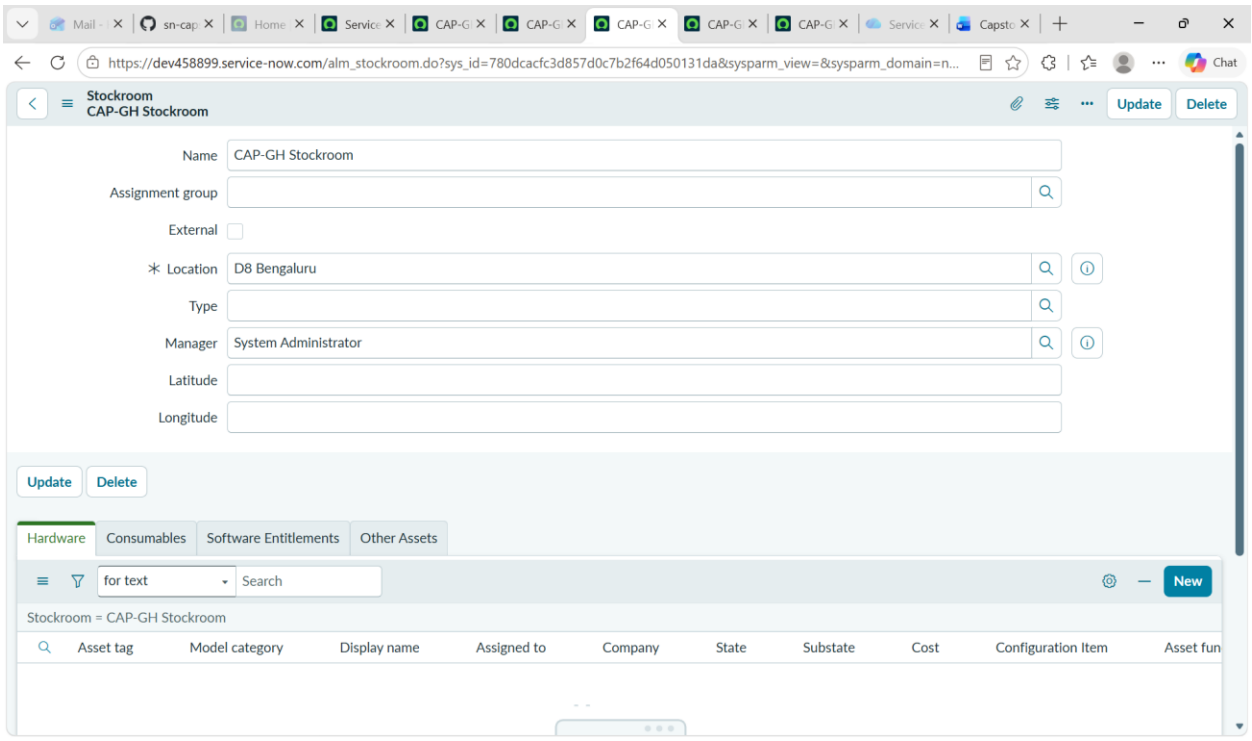
- C1-1a – Manufacturer record.



- C1-1b – Hardware model record.



- C1-2 – Stockroom record showing its manager.



C2 – Hardware Asset Creation

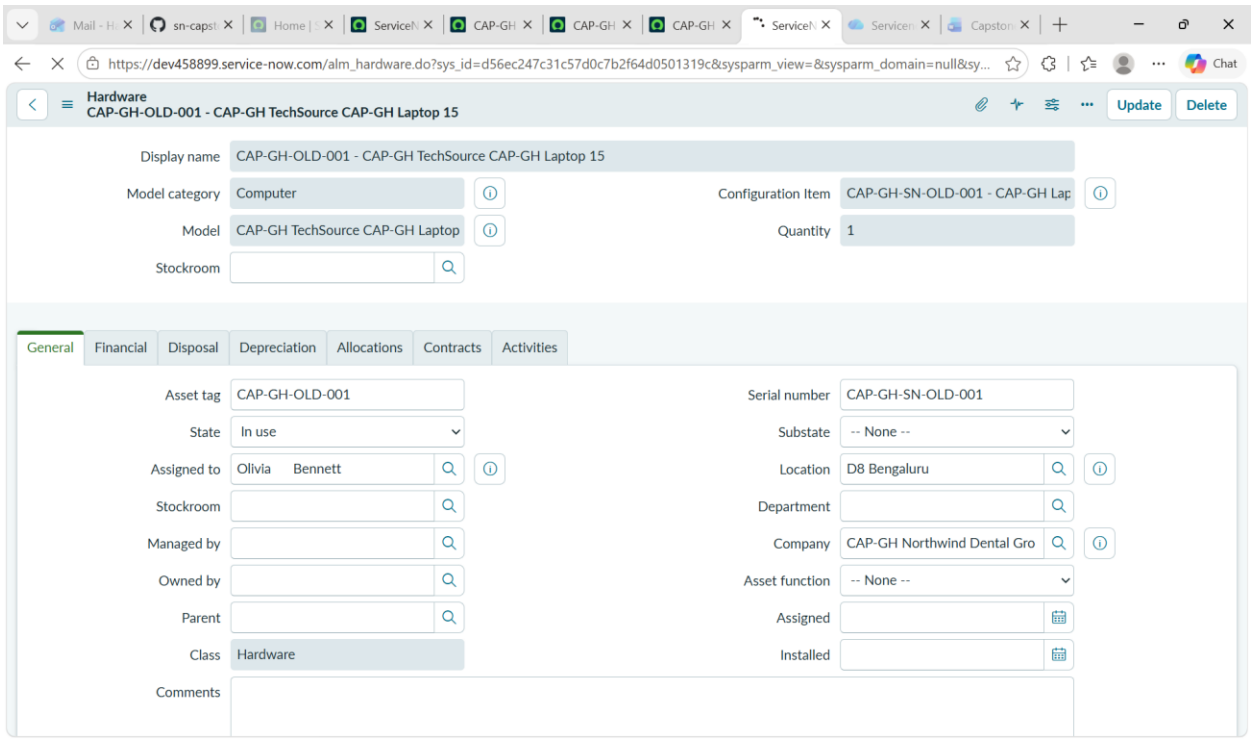
Created two hardware asset records using the hardware model **CAP-GH Laptop 15**.

1. **Faulty laptop**
    a. Asset tag: CAP-GH-OLD-001
    b. Serial number: CAP-GH-SN-OLD-001
    c. State: In use
    d. Assigned to: Olivia Bennett
2. **Replacement laptop**
    a. Asset tag: CAP-GH-NEW-001
    b. Serial number: CAP-GH-SN-NEW-001
    c. State: In stock
    d. Stockroom: CAP-GH Stockroom

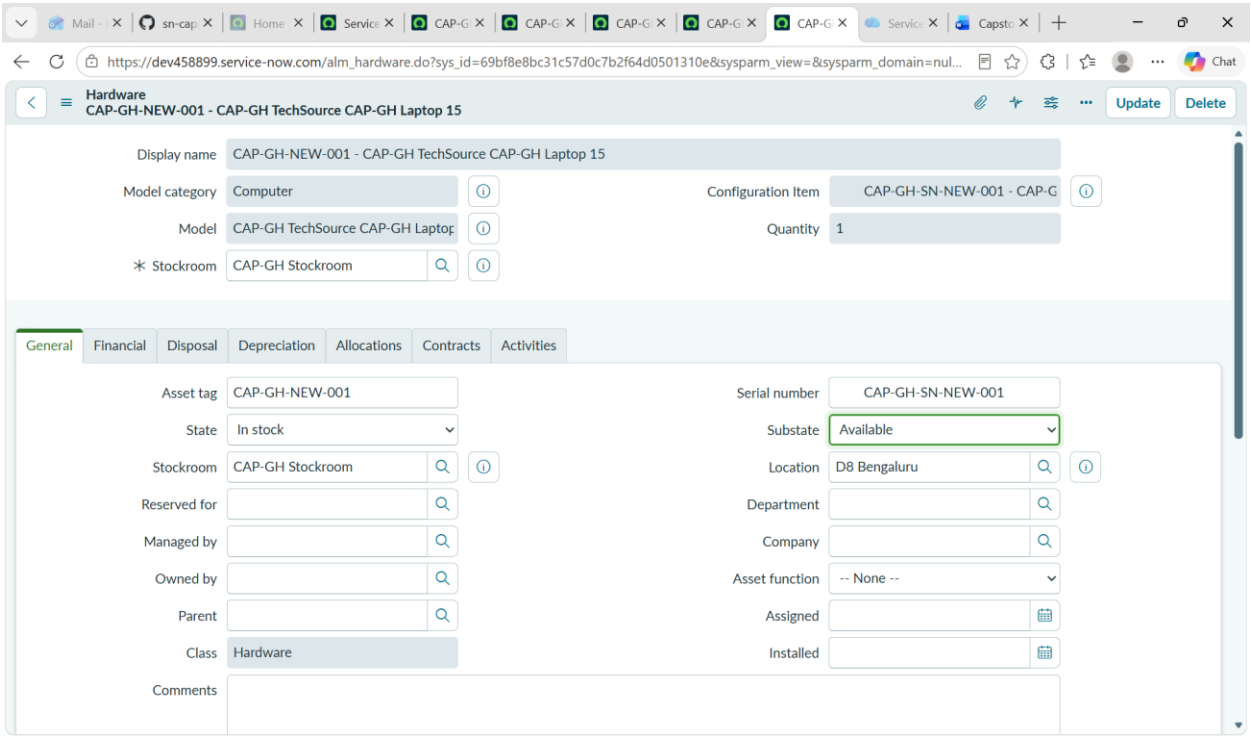
Both assets use the same hardware model, allowing the replacement device to be tracked through the subsequent field service and asset lifecycle processes.

Evidence:

- C2-2 – Faulty laptop asset record.



- C2-3 – Replacement laptop asset record.

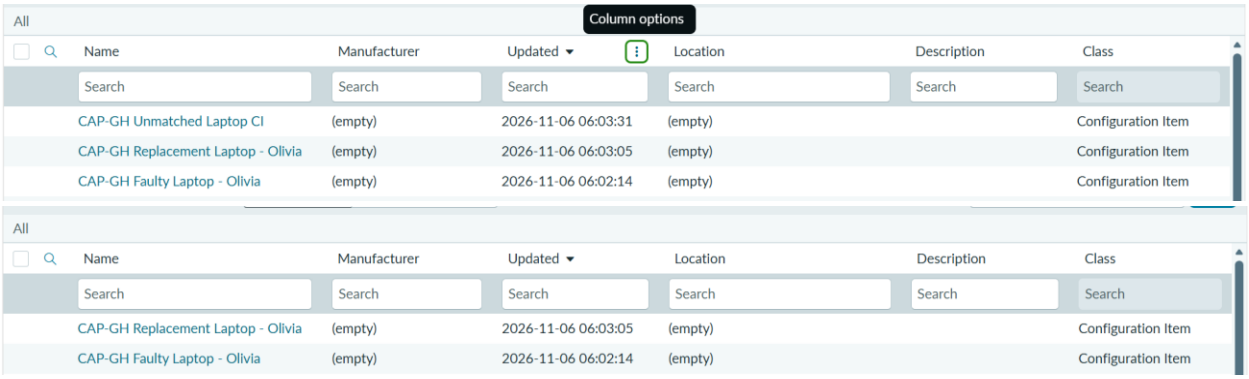


C3 – Configuration Item Linking and Reconciliation

Created two Configuration Items corresponding to the faulty and replacement laptops. Each CI was associated with its respective hardware asset using the asset tag as the reference.

A deliberate mismatch was introduced by creating an additional CI without an associated hardware asset. The mismatch was identified using a list filter that checked for CAP-GH Configuration Items with an empty Asset tag. The unmatched CI was removed, and the filter was executed again to verify the correction.

Evidence:



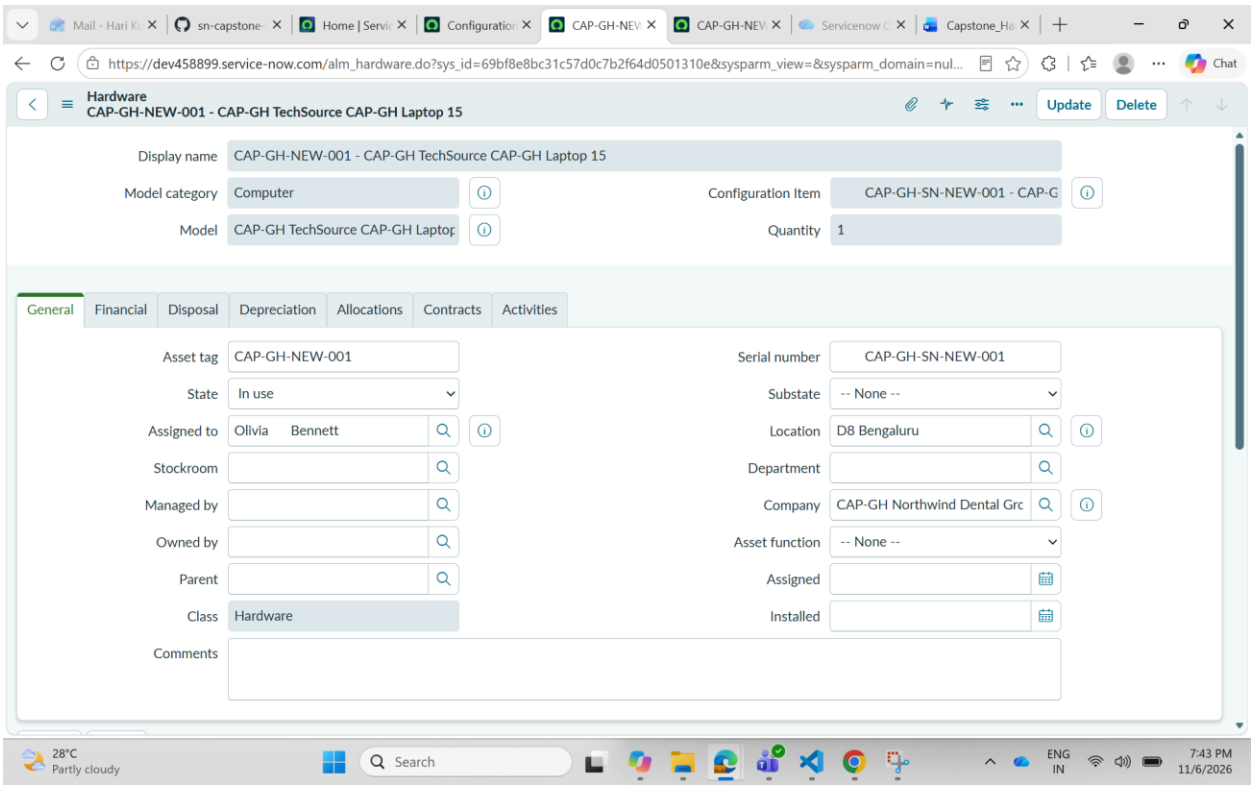
C4 – Replacement Asset Lifecycle

Following completion of the field service replacement activity, the replacement laptop was transitioned from In Stock to In Use and assigned to Olivia Bennett. Its stockroom assignment was cleared to reflect deployment.

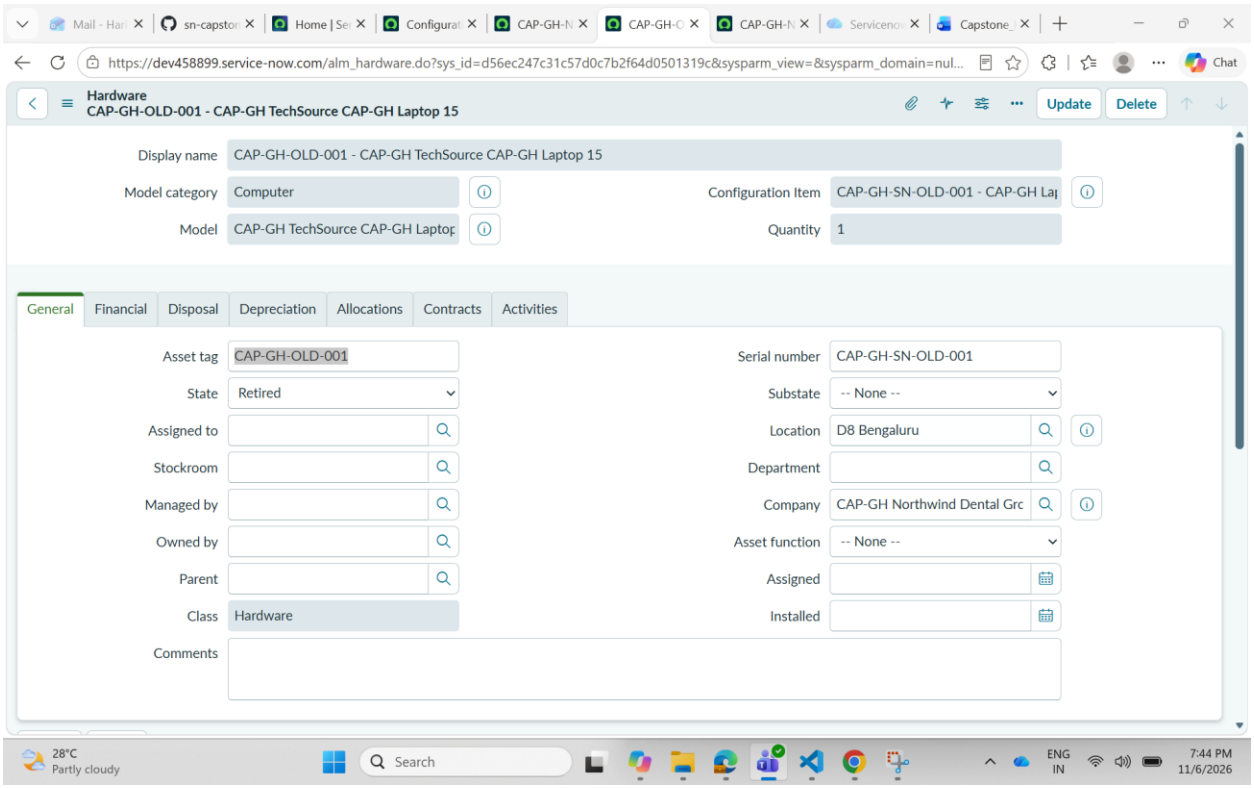
The faulty laptop was transitioned to the Retired state and its user assignment was removed. The asset history was reviewed to demonstrate the lifecycle changes and maintain traceability.

Evidence:

- C4-1 – Replacement laptop after deployment.



- C4-2 – Faulty laptop after retirement.



B1 – Work Order Creation from Customer Case

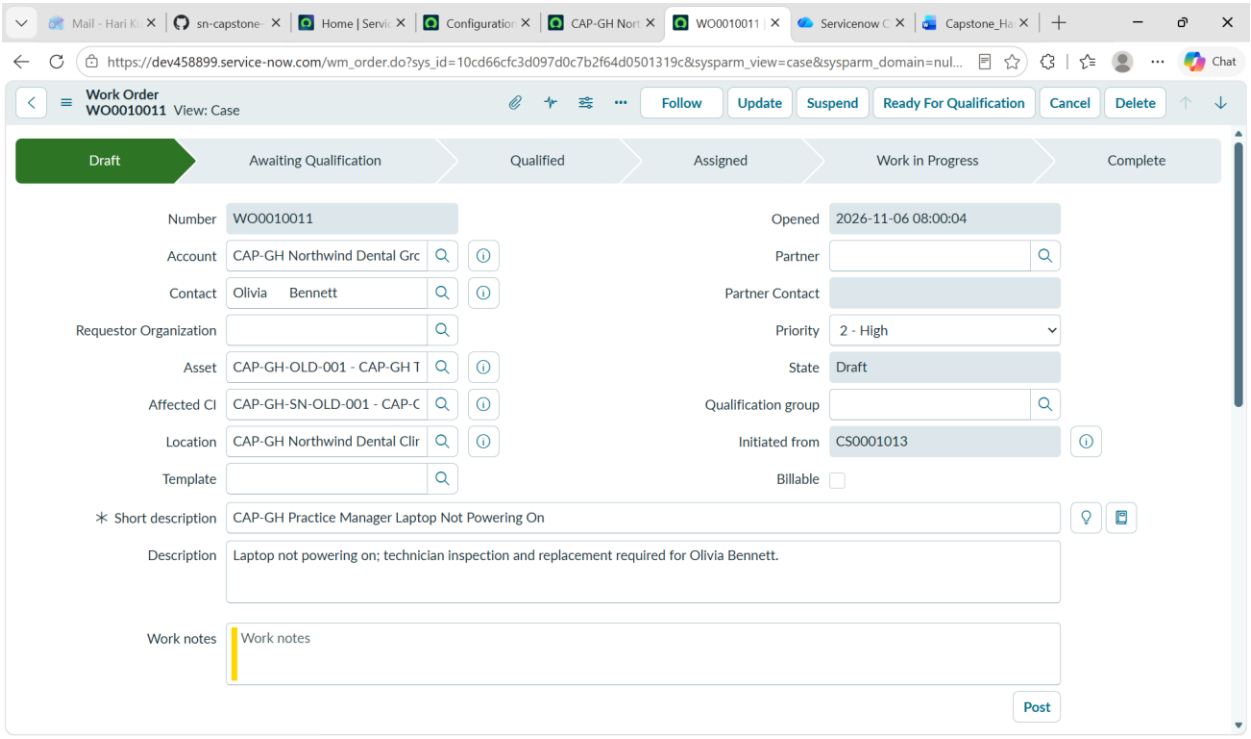
Created Field Service Work Order **WO0010011** directly from customer case **CS0001013**. ServiceNow automatically carried forward the customer account and contact information.

The work order is associated with CAP-GH Northwind Dental Group, contact Olivia Bennett, faulty hardware asset CAP-GH-OLD-001 and its corresponding Configuration Item CAP-GH-SN-OLD-001. The service location is CAP-GH Northwind Dental Clinic.

The work order description identifies the laptop power failure and the requirement for technician inspection and replacement. Priority was set to High (2).

The work order was initially created in Draft state and will be progressed through qualification and assignment during technician dispatch.

Evidence: B1-1 – Work order header and customer/device relationships.



B2 – Tasks, Skills and Dispatch

Two work order tasks were created under Field Service Work Order **WO0010011**.

1. Work Order Tasks

The following tasks were created:

Task	Task Number	Description
CAP-GH Diagnose laptop	WOT0010013	Inspect the faulty laptop, diagnose the power failure and record the findings.
CAP-GH Replace laptop and verify	WOT0010014	Replace the faulty laptop with the replacement device and verify its operation.

Both tasks reference the parent work order WO0010011 and the faulty laptop asset CAP-GH-OLD-001.

2. Task Dependency

A Finish-to-Start dependency was configured between the two tasks:

- Predecessor: WOT0010013 – CAP-GH Diagnose laptop.
- Successor: WOT0010014 – CAP-GH Replace laptop and verify.
- Dependency type: Finish to Start.
- Minimum lag time: Zero.

This ensures that the replacement task follows the diagnosis task.

3. Technician and Skill Configuration

A technician user, **Alex Technician**, was created with the user ID `cap_gh_technician` and the title Field Service Technician.

A custom skill named `CAP-GH` was created and associated with the technician at Basic skill level. The same skill was configured as a required skill for both work order tasks.

A dispatch group named **CAP-GH Field Service Dispatch** was created. It was configured with the `wm_dispatch` and `wm_work` roles, and Alex Technician was added as a group member.

Both tasks were assigned to Alex through this dispatch group and the corresponding assignment group. Scheduling was adjusted to avoid overlapping technician assignments.

4. Task Execution

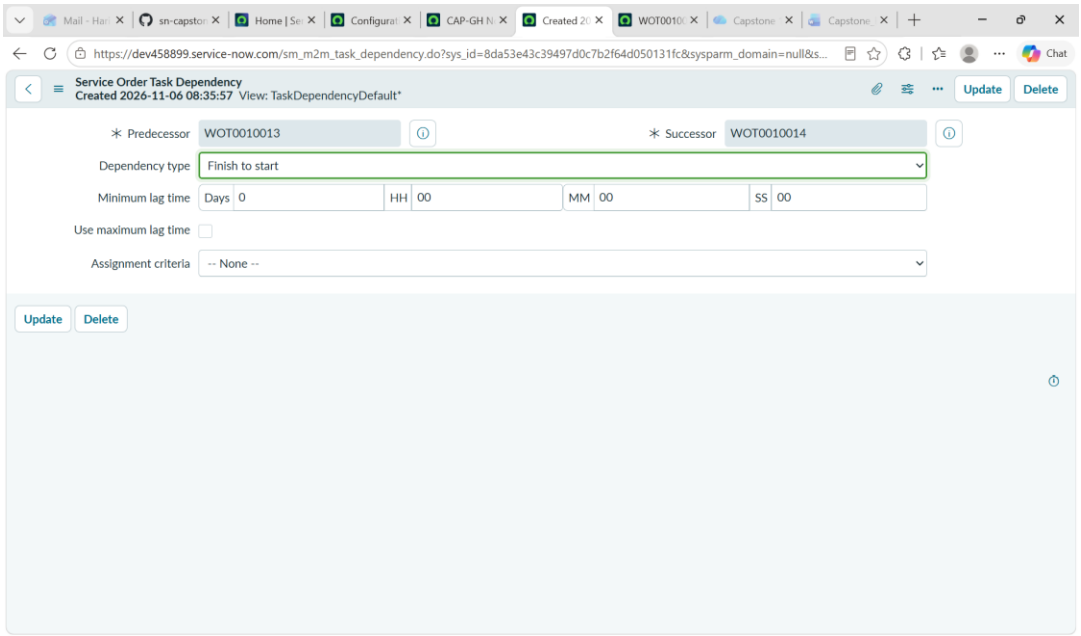
The technician tasks were completed in the required sequence:

1. The diagnosis task was completed first, recording the laptop inspection and fault diagnosis.
2. The replacement and verification task was completed after the diagnosis task.

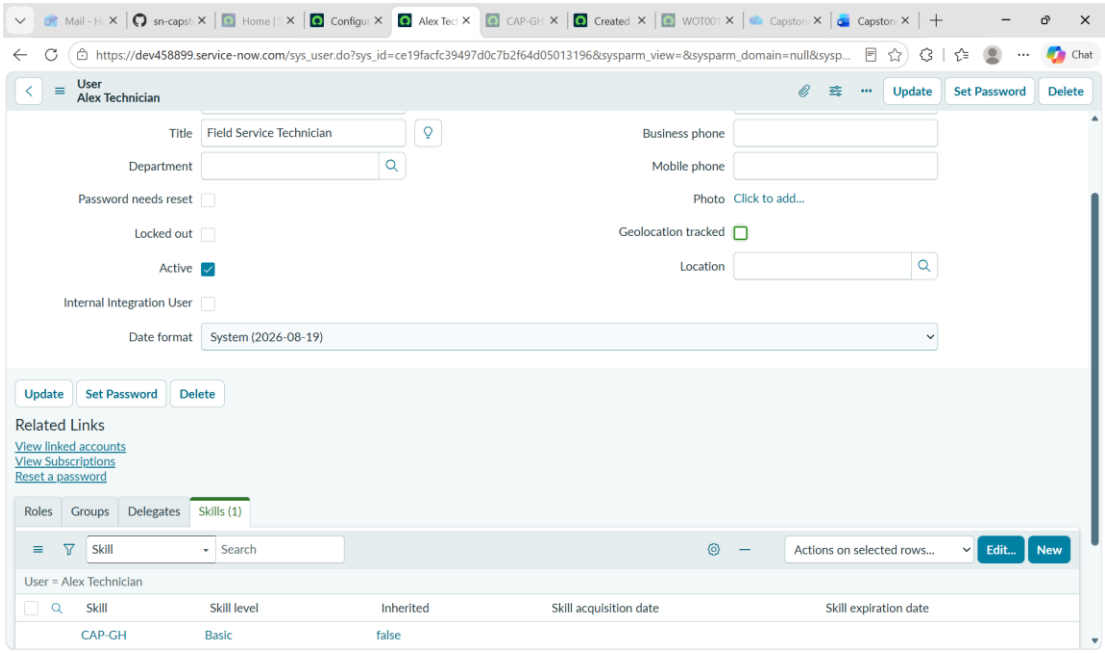
Both work order tasks reached the Completed state.

Evidence:

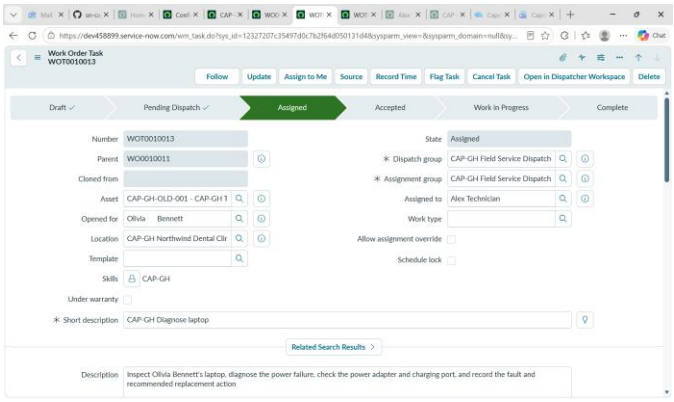
B2-1_Task_List_and_Dependency

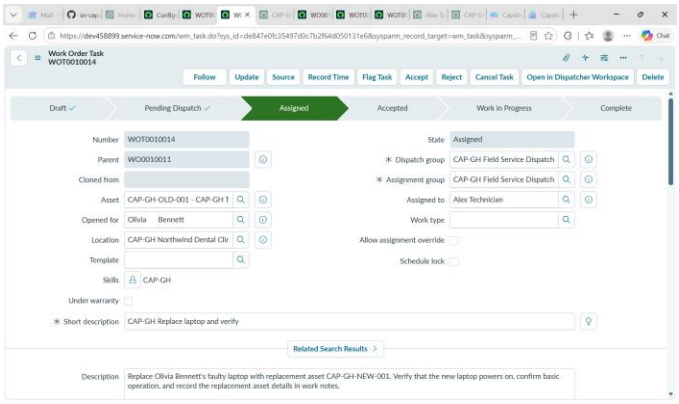


B2-2_Technician_Skills:

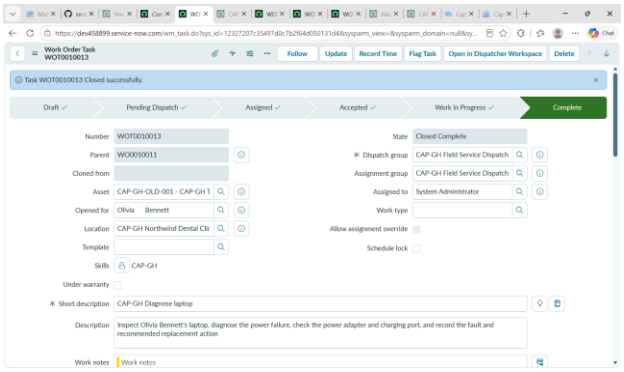


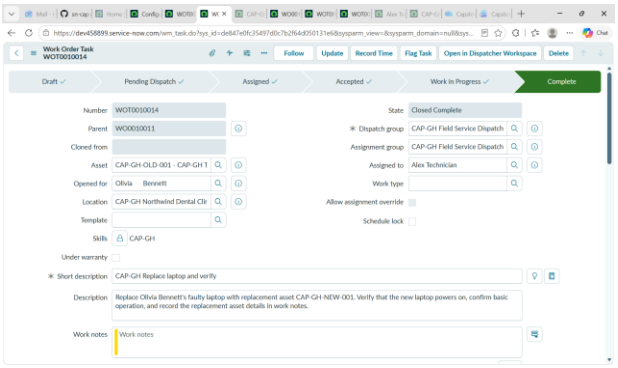
B2-3_Technician_Assignment:





B2-4_Completed_Tasks:





Outcome: The two dependent field service tasks were successfully created, linked, assigned to a technician with the required skill, and completed in the correct sequence. The parent work order completion is documented separately after its final state is verified.

B3 – Device on Work Order

The Field Service Work Order **WO0010011** was configured with the faulty laptop and its corresponding Configuration Item (CI).

1. Faulty Device Reference

The work order contains the following device references:

- Work Order: WO0010011
- Faulty Asset: CAP-GH-OLD-001
- Affected CI: CAP-GH-SN-OLD-001
- Customer: Olivia Bennett
- Related Case: CS0001013

The faulty laptop was linked through the Asset field, and its corresponding CI was selected in the Affected CI field.

2. Replacement Device Reference

After the replacement activity, the following information was recorded in the work order's Work notes:

- Faulty Asset: CAP-GH-OLD-001
- Faulty Device Serial Number: CAP-GH-SN-OLD-001
- Replacement Asset: CAP-GH-NEW-001
- Replacement Device Serial Number: CAP-GH-SN-NEW-001

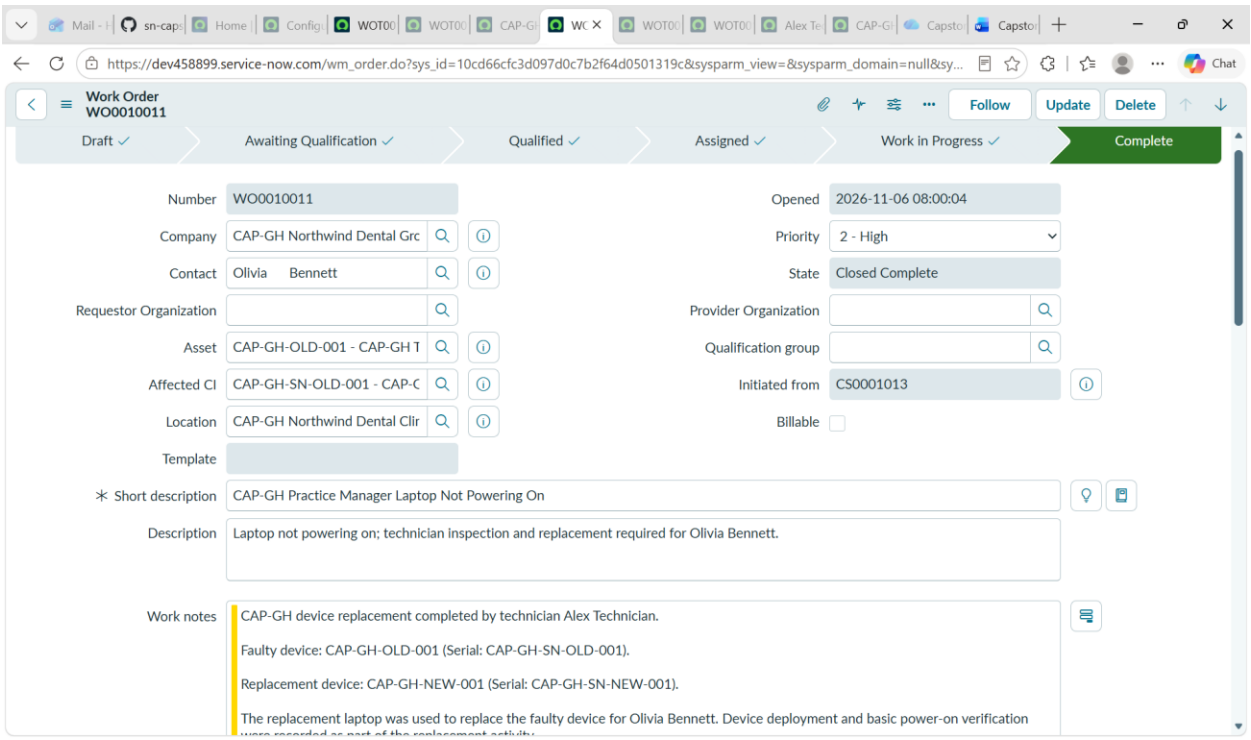
The work notes document the replacement of Olivia Bennett's faulty laptop with the designated replacement device and record the device deployment and basic power-on verification as part of the replacement activity.

3. Work Order Completion

The parent work order WO0010011 reached the **Closed Complete** state after its associated work order tasks were completed.

The work order therefore maintains the connection between the original customer case, the faulty asset, its affected CI, and the replacement device recorded in the work notes.

Evidence: B3-1_Device_Replacement_Reference



Outcome: The faulty device reference and replacement device details are documented in the Field Service Work Order, providing traceability for the laptop replacement process.

D1 – Software Installation Analysis

D1 – Software Installation Analysis

The Software Installations table was examined in the ServiceNow instance. Seven installation records were available.

The selected software product was **70400INVBUN_2010_0F**, published by **adskflex**, version **2010**.

Attribute	Observed Value
Software product	70400INVBUN_2010_0F
Publisher	adskflex

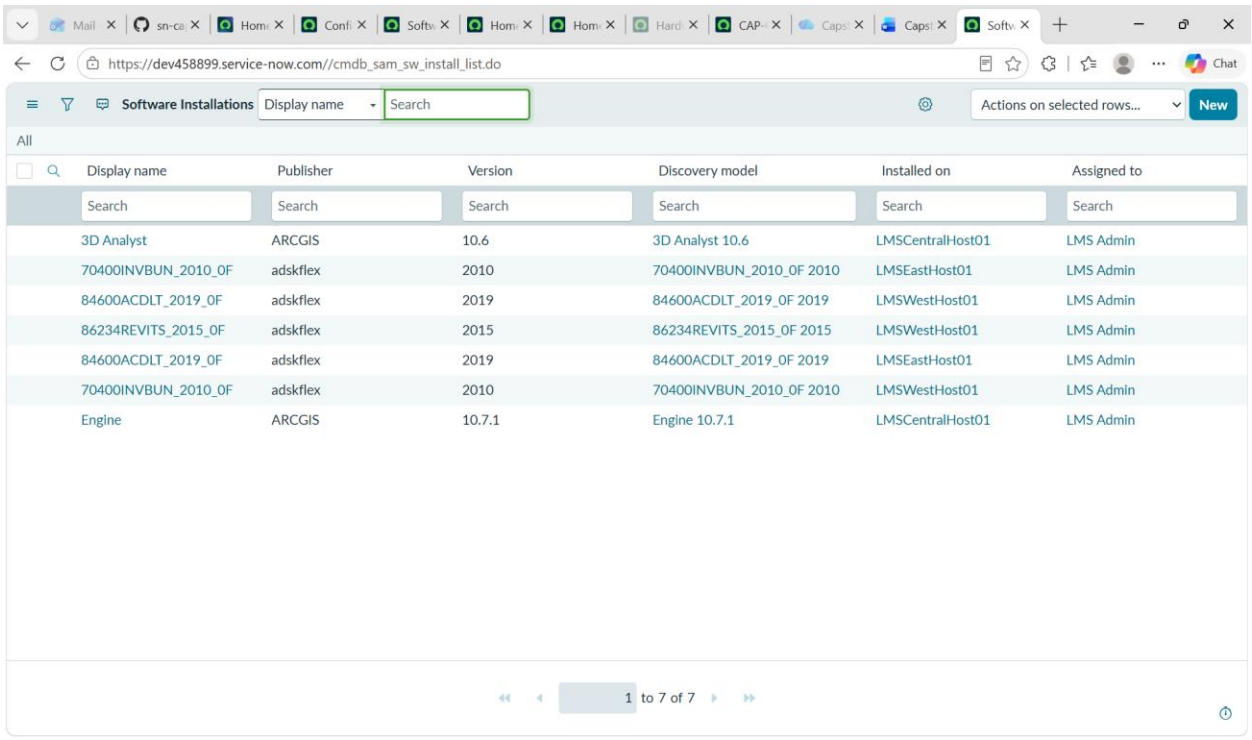
Version	2010
Total installations	2
Computer 1	LMSEastHost01
Computer 2	LMSWestHost01

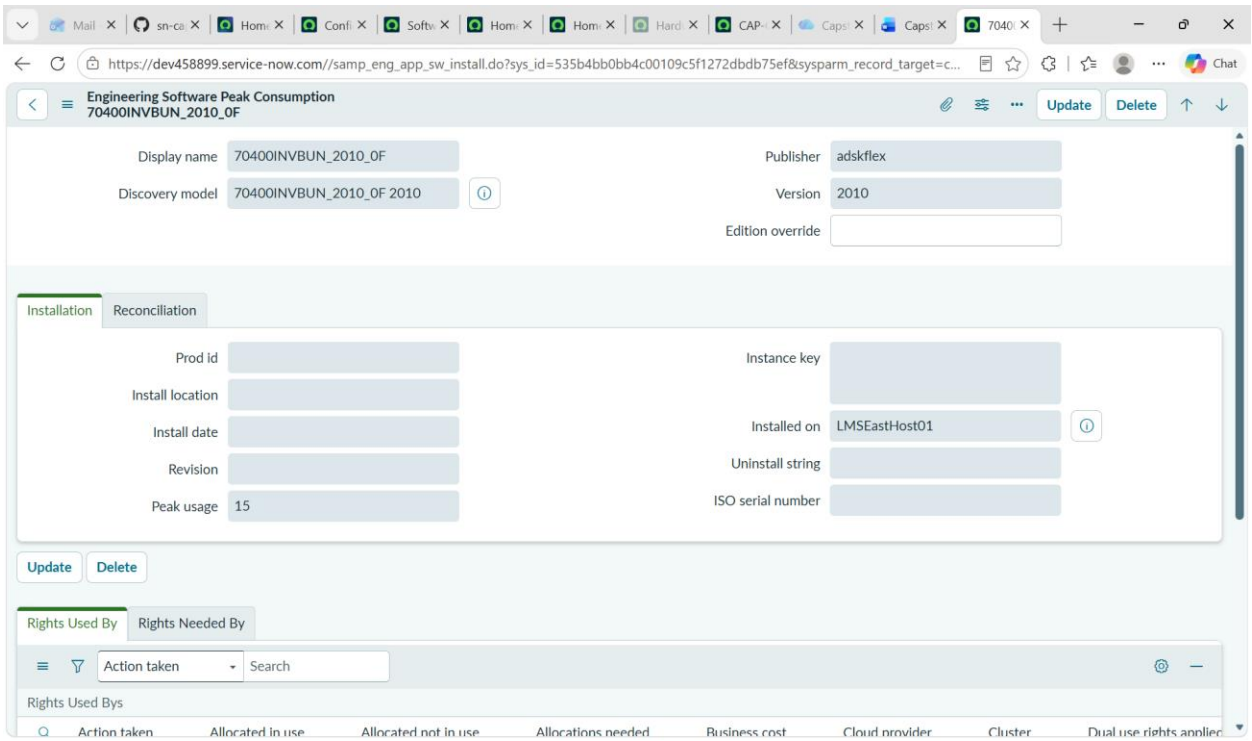
The selected product had two installation records across two computers. No individual product in the available installation list had five or more installations. The available data also did not provide three computers for a single product.

The selected product was retained for the licence entitlement analysis in D2 using the actual installation count.

Evidence:

D1-1_Software_Installations:


https://dev458899.service-now.com//cmdb_sam_sw_install_list.do
Software Installations
Display name
Search
Actions on selected rows...
New
All
Display name
Publisher
Version
Discovery model
Installed on
Assigned to
3D Analyst
ARCGIS
10.6
3D Analyst 10.6
LMSCentralHost01
LMS Admin
70400INVBUN_2010_0F
adskflex
2010
70400INVBUN_2010_0F 2010
LMSEastHost01
LMS Admin
84600ACDLT_2019_0F
adskflex
2019
84600ACDLT_2019_0F 2019
LMSWestHost01
LMS Admin
86234REVITS_2015_0F
adskflex
2015
86234REVITS_2015_0F 2015
LMSWestHost01
LMS Admin
84600ACDLT_2019_0F
adskflex
2019
84600ACDLT_2019_0F 2019
LMSEastHost01
LMS Admin
70400INVBUN_2010_0F
adskflex
2010
70400INVBUN_2010_0F 2010
LMSWestHost01
LMS Admin
Engine
ARCGIS
10.7.1
Engine 10.7.1
LMSCentralHost01
LMS Admin
1 to 7 of 7


https://dev458899.service-now.com//samp_eng_app_sw_install.do?sys_id=535b4bb0bb4c00109c5f1272dbdb75ef&sysparm_record_target=c...
Engineering Software Peak Consumption
70400INVBUN_2010_0F
Update
Delete
Display name
70400INVBUN_2010_0F
Publisher
adskflex
Discovery model
70400INVBUN_2010_0F 2010
Version
2010
Edition override
Installation
Reconciliation
Prod id
Instance key
Install location
Install date
Installed on
LMSEastHost01
Revision
Uninstall string
Peak usage
15
ISO serial number
Update
Delete
Rights Used By
Rights Needed By
Action taken
Search
Rights Used Bys
Action taken
Allocated in use
Allocated not in use
Allocations needed
Business cost
Cloud provider
Cluster
Dual use rights applied

D2 – Software Entitlement and Licence Recommendation

1. Software Model

A software model named **CAP-GH Inventor Professional 2010** was created in ServiceNow.

The selected software product is:

- **Product:** Inventor Professional
- **Publisher:** Autodesk Inc.
- **Version:** 2010
- **Product type:** Licensable
- **Status:** In Production

The model was created to represent the selected software product for licence entitlement management.

2. Software Entitlement

A software entitlement was created with the following configuration:

Field	Configured value
Display name	Autodesk Inc. Inventor Professional Per Device
Agreement type	Generic
License type	Perpetual
Metric group	Common
License metric	Per Device
Purchased rights	1
Active rights	1
Allocations available	1
Unit cost	100
Total cost	100
Initial state	Build

The entitlement was saved successfully. ServiceNow indicated that the entitlement must be published to change its state from **Build** to **In use**, allowing it to participate in reconciliation.

3. Licence Gap Calculation

The selected software has two installation records across two computers:

- LMSEastHost01
- LMSWestHost01

As the installation count is three or fewer, the required purchased rights are one.

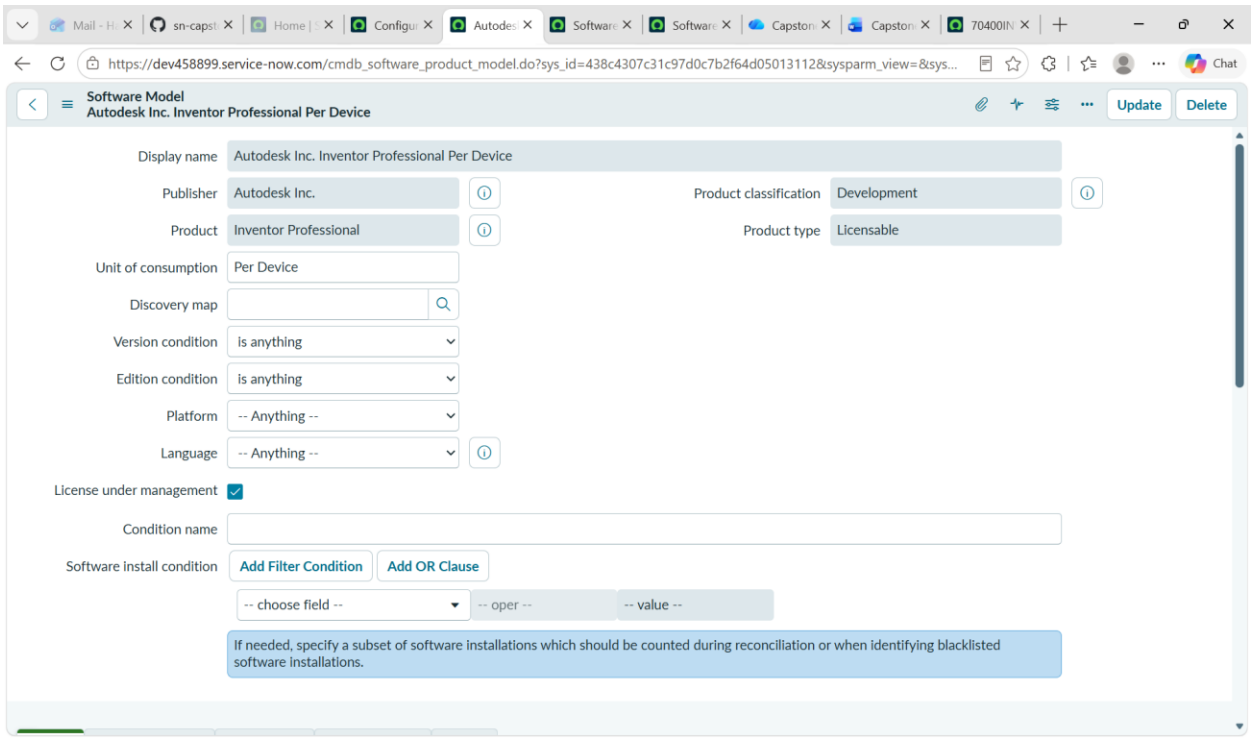
Calculation	Result
Total installations	2
Purchased rights	1
Licence gap	2 − 1 = +1
Licence position	Under-licensed
Assumed cost per right	100
Cost to close the gap	1 × 100 = 100

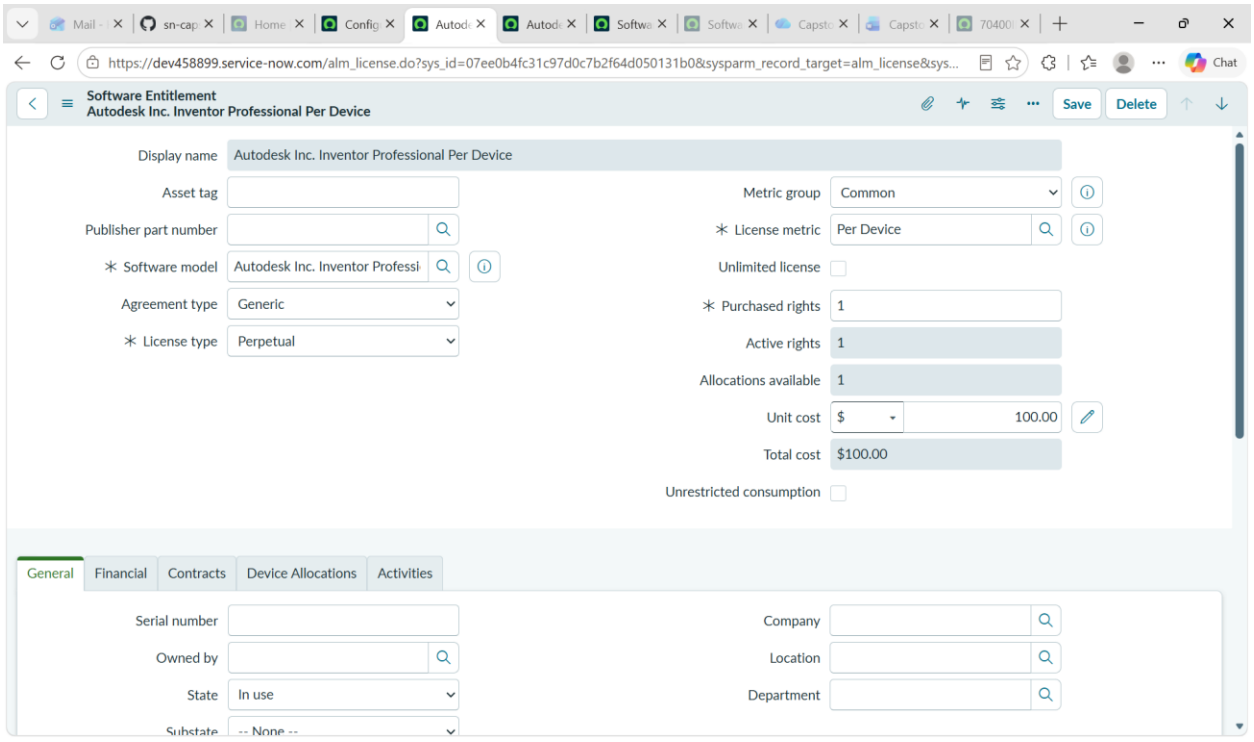
4. Recommendation

The software is under-licensed by one device licence. The recommended action is to purchase one additional Per Device licence at the assumed cost of 100, bringing the total purchased rights to two.

An alternative is to review the two installation records and identify whether either installation is unused or no longer required. If an installation can be safely removed and its licence reclaimed, the organisation may avoid purchasing an additional licence.

Evidence:


https://dev458899.service-now.com/cmdb_software_product_model.do?sys_id=438c4307c31c97d0c7b2f64d05013112&sysparm_view=&sys...
Chat
Software Model
Autodesk Inc. Inventor Professional Per Device
Update
Delete
Display name
Autodesk Inc. Inventor Professional Per Device
Publisher
Autodesk Inc.
Product
Inventor Professional
Unit of consumption
Per Device
Discovery map
Version condition
is anything
Edition condition
is anything
Platform
-- Anything --
Language
-- Anything --
License under management
Condition name
Software install condition
Add Filter Condition
Add OR Clause
-- choose field --
-- oper --
-- value --
If needed, specify a subset of software installations which should be counted during reconciliation or when identifying blacklisted software installations.
Product classification
Development
Product type
Licensable


https://dev458899.service-now.com/alm_license.do?sys_id=07ee0b4fc31c97d0c7b2f64d050131b0&sysparm_record_target=alm_license&sys...
Chat
Software Entitlement
Autodesk Inc. Inventor Professional Per Device
Save
Delete
Display name
Autodesk Inc. Inventor Professional Per Device
Asset tag
Publisher part number
Software model
Autodesk Inc. Inventor Professi
Agreement type
Generic
License type
Perpetual
Metric group
Common
License metric
Per Device
Unlimited license
Purchased rights
1
Active rights
1
Allocations available
1
Unit cost
$
100.00
Total cost
$100.00
Unrestricted consumption
General
Financial
Contracts
Device Allocations
Activities
Serial number
Owned by
State
In use
Substate
-- None --
Company
Location
Department

5. Outcome

The software model and entitlement record were created, and the entitlement's licence metric, purchased rights and cost were configured. The calculated licence position is under-licensed by one right, with an estimated cost of 100 to close the gap.

E1 – End-to-End Traceability

The following table summarises the records created during the laptop replacement scenario and the relationships connecting the CSM, FSM and ITAM processes.

Module	Record	Identifier / Name	Relationship
CSM	Customer Contact	Olivia Bennett	Contact associated with Northwind Dental Group
CSM	Customer Case	CS0001013	Reports the laptop power failure
FSM	Work Order	WO0010011	Initiated from case CS0001013
FSM	Diagnosis Task	WOT0010013	Task under WO0010011
FSM	Replacement Task	WOT0010014	Depends on WOT0010013
ITAM	Faulty Asset	CAP-GH-OLD-001	Laptop originally assigned to Olivia
ITAM	Replacement Asset	CAP-GH-NEW-001	Replacement laptop deployed to Olivia
ITAM / CMDB	Faulty CI	CAP-GH Faulty Laptop - Olivia	Identified using the faulty asset tag
ITAM / CMDB	Replacement CI	CAP-GH Replacement Laptop - Olivia	Identified using the replacement asset tag
SAM	Software Product	70400INVBUN_2010_0F – Inventor Professional 2010	Software selected for installation and licence analysis

The case initiated the field service work order, which contained two dependent tasks for diagnosis and replacement. The asset records were updated to reflect the device exchange, while the corresponding configuration items were created for tracking. Software installation and entitlement analysis was also performed as part of the IT asset management process.

E2 – Project Reflection

This project demonstrated how ServiceNow connects Customer Service Management (CSM), Field Service Management (FSM) and IT Asset Management (ITAM) in an end-to-end IT service workflow.

The process began when Olivia Bennett reported that her laptop was not powering on. A CSM case was created to record the issue and associate it with the customer account and support information. The case was then used to initiate a field service work order.

The work order contained two dependent tasks: diagnosing the faulty laptop and replacing the device. Completing these tasks in the correct sequence supported the replacement process. The faulty laptop asset was retired, and the replacement laptop was deployed to the customer. Corresponding Configuration Items were also created to support configuration tracking.

Finally, software installations were examined and a software entitlement was configured to analyse licence availability. This demonstrated how software asset information can help identify licensing gaps and support purchasing or licence reclamation decisions.

Suggested Improvements

1. **Automated workflows:** Improve automation between case creation, work order generation and task assignment to reduce manual effort.
2. **Hardware Asset Management Pro:** Introduce lifecycle automation to improve asset tracking, transfers, replacement and retirement processes.
3. **Software Asset Management Pro:** Use software reclamation and reconciliation capabilities to identify unused licences, reduce unnecessary purchases and improve licence compliance.

Overall, this project provided practical experience in connecting customer support, field operations and IT asset management through a single ServiceNow platform.